

Cooperative Multi-Agent Reinforcement Learning for Energy-Aware UPF Realisation Switching at the 5G Edge

Shima Afshar Borji*, Roberto Bruschi, Chiara Lombardo, Raffaele Bolla, and Cristina Emilia Costa

Abstract—5G User Plane Function instances at the edge can run as DPDK kernel-bypass (high throughput, high idle power) or as user-space (lower power, narrower envelope). We cast per-site switching across $K = 10$ heterogeneous edge clusters as cooperative reinforcement learning with a reward that folds the digital twin’s measured switching energy into specific energy consumption, removing the parallel flat-cost term used in prior work. On a 10.5-day held-out test slice, MAPPO attains -5633 ± 100 reward over 8 seeds, beating an independent-PPO ensemble (-5787 ± 89 ; paired-bootstrap 95 % CI $[+28, +262]$, $p = 0.009$), centralised single-MLP PPO, and the swept-best classical hysteresis baseline by 10–61 %.

Index Terms—5G User Plane Function, multi-agent reinforcement learning, MAPPO, centralised training decentralised execution, energy efficiency, mobile edge computing.

I. INTRODUCTION

THE 5G core’s User Plane Function (UPF) is the highest-throughput data-plane element in a typical edge deployment. Two realisations are common: a DPDK-accelerated kernel-bypass stack (high throughput, high idle power) and a user-space stack (lower power, narrower operating envelope). Static all-DPDK wastes energy at low load; static all-USR violates QoS at peaks. The right realisation depends on the per-site forecast and on a non-trivial switching cost. Coordinating this choice across many edge clusters is a cooperative control problem whose joint policy space scales as 2^K for K clusters.

Prior work [1] formulates the per-site reward with a flat switching-cost constant ($c_{\text{DPDK}}, c_{\text{USR}}$) running in parallel to a digital twin that already exposes the measured per-transition spike energy $E_{i,t}^{\text{sw}}$. We remove that redundancy: the spike is folded into specific energy consumption (SEC) so a single physical quantity drives the energy term. Our contributions are:

- 1) A unified physics-grounded reward in which switching cost emerges from measured activation power $\times$ duration rather than a hand-set constant (Section II).
- 2) A controlled three-architecture comparison at fixed budget and eight training seeds—centralised single-MLP PPO, an independent-PPO ensemble (IPPO), and

MAPPO [2] with a parameter-shared actor and a centralised critic—against three classical baselines (always-DPDK; derived-threshold; hysteresis swept over five band widths) (Sections III–IV).

- 3) Statistical evidence that the MAPPO architectural win over IPPO is significant under the revised reward: paired bootstrap $p = 0.009$, 95 % CI $[+28, +262]$ reward units; 7 of 8 seeds favour MAPPO (Section IV).

II. SYSTEM MODEL AND REWARD

A mobile operator deploys $K = 10$ MEC clusters (mean offered loads 0.108–1.808 Gbps; peaks to 5.58 Gbps). At each control step t (15 min), cluster i picks $a_{i,t} \in \{0, 1\}$ (0 = DPDK, 1 = USR). A calibrated digital twin [6] returns, per cluster and realisation, the steady-state power $P_{i,t}^{\text{steady}}$, delay $d_{i,t}$, predicted loss $\ell_{i,t}$, and an `is_safe` flag. Per-cluster loads come from an upstream forecaster [5] partitioned chronologically into train (5073 steps ≈ 53 d), validation (1009 steps), and test (1009 steps). Forecaster MAE on test averaged over K clusters is 110.7 Mbps.

Each agent’s 14-dim observation comprises the current load, an 8-step history window, the 1-step forecast, the previous action, the previous QoS score, the previous SEC, and a normalised cooldown progress. The cooperative objective is the discounted sum of per-cluster rewards over $T = 1009$ steps with $\gamma = 0.995$.

Reward (physics-grounded). With $\Delta t = 900$ s and the twin’s per-transition spike energy $E_{i,t}^{\text{sw}} = P_{\text{new}}(\lambda_{i,t}) t_{\text{new}}^{\text{act}}$ (activation durations: $t_{\text{DPDK}}^{\text{act}} = 24$ s, $t_{\text{USR}}^{\text{act}} = 3.3$ s, from RAPL measurements [6]), define

$$P_{i,t}^{\text{eff}} = P_{i,t}^{\text{steady}} + E_{i,t}^{\text{sw}} / \Delta t, \quad (1)$$

$$\text{SEC}_{i,t} = P_{i,t}^{\text{eff}} / \max(\lambda_{i,t}, \varepsilon). \quad (2)$$

The per-step reward is

$$r_{i,t} = -(\alpha \text{SEC}_{i,t} + \lambda_{\text{QoS}} \max(0, \tau - Q_{i,t}) + L_{i,t}^{\text{CD}}), \quad (3)$$

where $Q_{i,t} \in [0, 1]$ scores delay and loss against budgets, $L_{i,t}^{\text{CD}}$ is a soft anti-oscillation penalty ($c_{\text{CD}} = 0.5$, period 4 steps), and $(\alpha, \lambda_{\text{QoS}}, \tau) = (100, 30, 0.9)$ are paper-aligned [1]. The asymmetry between DPDK and USR switching costs emerges automatically from $t_{\text{DPDK}}^{\text{act}} \gg t_{\text{USR}}^{\text{act}}$ and from load scaling, not from a hand-set constant.

Classical baselines. The threshold and hysteresis policies use operating points derived from the twin’s measured break-even (91.0 Mbps where USR power $\geq$ DPDK) and QoS

S. Afshar Borji, R. Bruschi, C. Lombardo, and R. Bolla are with DITEN, University of Genoa, Genoa, Italy (e-mail: shima.afshar.borji@edu.unige.it; roberto.bruschi@unige.it; chiara.lombardo@unige.it; raffaele.bolla@unige.it).

R. Bruschi, C. Lombardo, R. Bolla, and C. E. Costa are also with CNIT – S2N National Lab, Genoa, Italy (e-mail: ccosta@cnit.it).

*Corresponding author.

ORCIDs to be inserted at submission.

TABLE I
TEST-SLICE COMPARISON ($K=10$, 1009-STEP EPISODE). MAPPO AND IPPO AT 8 SEEDS; CENTRALISED PPO AT 4. REWARD IS THE UNWEIGHTED SUM OVER CLUSTERS AND STEPS.

Controller	Reward n	Wh	Uns%
MAPPO (ours)	-5633 ± 100	8 1967	0.52
IPPO ensemble	-5787 ± 89	8 1973	0.57
Hysteresis ($b=50$)	$-6262 -$	1995	0.68
Hysteresis ($b=20$)	$-6543 -$	1971	0.94
Always DDPK	$-7201 -$	2071	0.38
Hysteresis (auto b)	$-7201 -$	2071	0.38
Threshold 81 Mbps	$-7623 -$	1970	1.42
Centralised PPO	$-14\,458 \pm 4509$	4 2130	3.73

limit (149.0 Mbps where USR is still safe), giving a decision threshold of 81 Mbps after a 10 Mbps safety margin. The hysteresis band is swept over $\{5, 10, 20, 50, 100\}$ Mbps with cooldown of one step.

III. METHOD

MAPPO [2] uses a parameter-shared actor (MLP $14 \rightarrow 64 \rightarrow 64 \rightarrow 2$, 5 250 parameters total—one network handles every agent’s per-step decision) and a centralised critic on the global state (MLP $140 \rightarrow 128 \rightarrow 128 \rightarrow K$, 35 850 parameters). Training uses PPO clipping [3] on per-agent ratios with GAE [4] ($\lambda = 0.9$), per-agent advantage normalisation over (T, K) , reward scaling 0.01 to balance actor and critic losses, 200k env steps, $n_{\text{steps}} = 1024$, $n_{\text{epochs}} = 10$, $\text{lr} \ 3 \times 10^{-4}$, entropy coefficient 0.05. Best-of-validation checkpoint selection is load-bearing: without it, late-stage policy collapse on a near-zero-load step produces an unusable final-step model.

IPPO ensemble trains $K = 10$ independent Stable-Baselines3 [7] PPO actors, one per cluster, on each cluster’s 14-dim observation with identical hyperparameters. At evaluation, the per-cluster policies are stacked.

Centralised PPO is a single SB3 PPO on the joint 140-dim observation with a MultiDiscrete($[2]^K$) action space, otherwise matched. Each trained controller is evaluated once on the test slice with $\text{seed}=42$ for environment reset and a deterministic policy. MAPPO and IPPO are run at eight seeds $\{1, 7, 13, 23, 42, 64, 77, 99\}$; centralised PPO at four seeds.

IV. RESULTS

Table I reports test-slice means across all controllers. MAPPO attains the lowest reward, energy, and unsafe rate among trained policies. The gap to the swept-best classical baseline (hysteresis, band = 50 Mbps) is 629 reward units; the gap to always-DDPK is 1568 units. Centralised single-MLP PPO collapses (seed range $\sim 9\,930$) because the joint factored softmax cannot factor the per-cluster sub-structure within the training budget; its mean is worse than always-DDPK.

Statistical significance. A paired bootstrap (10 000 resamples of seed-matched MAPPO – IPPO deltas) places the architectural gap at +154 reward units, 95 % CI $[+28, +262]$, one-sided $p=0.009$. Seven of eight seeds favour MAPPO (Fig. 1); the only counter-example (seed 13) is the seed where MAPPO

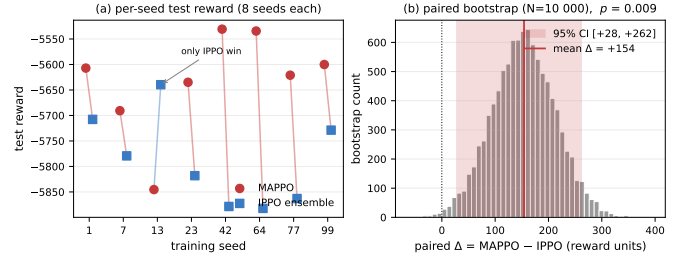


Fig. 1. (a) Per-seed test reward for MAPPO (circles) and the IPPO ensemble (squares) across eight seeds; paired lines connect seed-matched runs. Seed 13 is the only seed where IPPO wins. (b) Paired-bootstrap distribution (10 000 resamples) of the mean $\Delta = \text{MAPPO} - \text{IPPO}$ delta; the 95 % CI $[+28, +262]$ excludes zero ($p = 0.009$).

converged to a slightly more aggressive USR-usage policy (0.67 % unsafe, 364 flips vs the average 0.52 %, 290 flips).

Per-cluster transfer. MAPPO’s win concentrates on the lowest-load cluster (c0, mean load 0.108 Gbps, +6.6 %) and on mid-load clusters where the IPPO ensemble’s per-cluster PPO failed to leave always-DDPK because per-cluster data alone was insufficient signal (c5 +8.6 %, c7 +4.3 %, c4 +3.4 %). MAPPO’s shared actor—trained simultaneously on the data-rich clusters c0/c1/c4/c6/c7—generalises the “USR-at-low-predicted-load” rule to c5, where IPPO never escapes DDPK. MAPPO wins on 7 of 10 clusters, ties on 1, and loses by $\leq 9\%$ on the remaining 2.

Classical-baseline sensitivity. Sweeping the hysteresis band over $\{5, 10, 20, 50, 100\}$ Mbps identifies band=50 as the strongest classical operating point (−6262); band ≤ 10 over-switches (unsafe $> 1.1\%$); band=100 collapses to always-DDPK because $t_{\text{down}} \rightarrow 0$. *No setting of the band closes the gap to MAPPO.* A complementary cooldown sweep on MAPPO around the default (period = 4, cost = 0.5) confirms the result is not load-bearing on the cooldown hyperparameters (full sweep tables in the supplementary material).

V. CONCLUSION

Per-site DDPK/USR realisation switching across heterogeneous edge clusters is a cooperative-MARL problem whose architectural choice matters more than the algorithm. Under a physics-grounded reward that unifies steady-state and switching energy, MAPPO with a shared-parameter actor and centralised critic statistically beats both an independent-PPO ensemble ($p = 0.009$) and the swept-best classical hysteresis controller (629 reward units) on a 10.5-day held-out test slice from a calibrated digital twin. The advantage concentrates on data-poor mid-load clusters, consistent with cross-cluster policy transfer through the shared actor. Trained checkpoints, the digital twin, and an interactive dashboard are released for reproducibility.¹

REFERENCES

- [1] [TO VERIFY] Source paper for the paper-aligned reward shape (SEC + asymmetric switching cost + QoS penalty), COMCOM-S-26-00430.

¹Public artefact URL to be inserted at submission.

- [2] C. Yu, A. Velu, E. Vinitzky, J. Gao, Y. Wang, A. Bayen, and Y. Wu, “The surprising effectiveness of PPO in cooperative multi-agent games,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2022.
- [3] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, “Proximal policy optimization algorithms,” *arXiv preprint arXiv:1707.06347*, 2017.
- [4] J. Schulman, P. Moritz, S. Levine, M. Jordan, and P. Abbeel, “High-dimensional continuous control using generalized advantage estimation,” in *Int. Conf. Learn. Represent. (ICLR)*, 2016.
- [5] [TO VERIFY] Upstream traffic forecaster (UpfTrafficForecaster) producing per-cluster predictions used in this work.
- [6] [TO VERIFY] Upstream digital-twin and threshold-derivation framework (UPF_NDT); also the source of the RAPL-measured DPDK/USR activation durations.
- [7] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, and N. Dornmann, “Stable-Baselines3: Reliable reinforcement learning implementations,” *J. Mach. Learn. Res.*, vol. 22, no. 268, pp. 1–8, 2021.